React Capstone Assignment

1. Project Summary

The main goal of my React capstone project is to build an AI-Powered Playlist Creator that helps users create personalized music playlists based on their mood, favorite artists, genres, or activities. Users can log in, search for songs, receive AI-based playlist recommendations, save playlists, and listen to music through an easy-to-use interface.

2. Five Core Features

1. User Login & Authentication – Secure access to personal playlists.
2. AI Playlist Recommendation – Suggests playlists based on mood.
3. Song Search – Helps users quickly find songs and artists.
4. Create & Save Playlists – Lets users organize favorite songs.
5. Recently Played & Favorites – Gives quick access to frequently played songs.

3. Flow Diagram

Start → Login/Sign Up → Home Dashboard → Search Song or Select Mood → AI Generates Playlist → Preview Playlist → Save Playlist → Play Music → Add to Favorites → Logout.

ChatGPT suggested adding mood-based playlist recommendations and a Recently Played section.

4. React Implementation

Feature: AI Playlist Recommendation.
Components: MoodSelector and PlaylistResults.
Shared data: selectedMood (string) and playlist (array).
Example state:

```
const [selectedMood, setSelectedMood] = useState('');
const [playlist, setPlaylist] = useState([]);
```

MoodSelector updates the selected mood, App stores the state, and PlaylistResults displays the recommended playlist.